

SERVICES, DIGITAL ECONOMY, FINTECH AND PLATFORM MARKETS

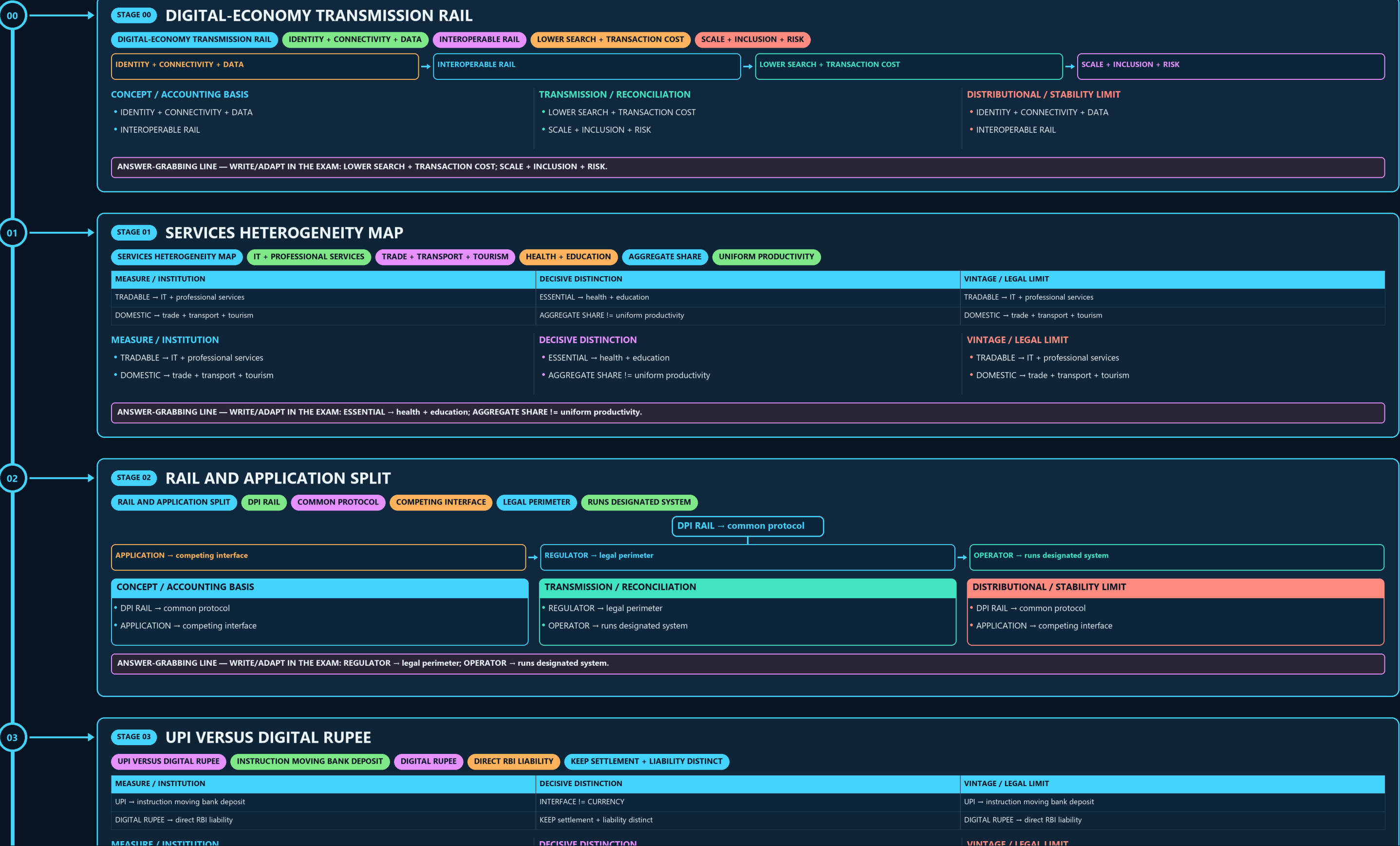
DIGITAL-ECONOMY TRANSMISSION RAIL → SERVICES HETEROGENEITY MAP → RAIL AND APPLICATION SPLIT → UPI VERSUS DIGITAL RUPEE → ONDC INTEROPERABILITY CHAIN → PLATFORM POWER LOOP

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



- DPI RAIL → common protocol
- APPLICATION → competing interface

- REGULATOR → legal perimeter
- OPERATOR → runs designated system

- DPI RAIL → common protocol
- APPLICATION → competing interface

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REGULATOR → legal perimeter; OPERATOR → runs designated system.

03

STAGE 03 UPI VERSUS DIGITAL RUPEE

UPI VERSUS DIGITAL RUPEE INSTRUCTION MOVING BANK DEPOSIT DIGITAL RUPEE DIRECT RBI LIABILITY KEEP SETTLEMENT + LIABILITY DISTINCT

| MEASURE / INSTITUTION                 | DECISIVE DISTINCTION                 | VINTAGE / LEGAL LIMIT                 |
|---------------------------------------|--------------------------------------|---------------------------------------|
| UPI → instruction moving bank deposit | INTERFACE != CURRENCY                | UPI → instruction moving bank deposit |
| DIGITAL RUPEE → direct RBI liability  | KEEP settlement + liability distinct | DIGITAL RUPEE → direct RBI liability  |

MEASURE / INSTITUTION

- UPI → instruction moving bank deposit
- DIGITAL RUPEE → direct RBI liability

DECISIVE DISTINCTION

- INTERFACE != CURRENCY
- KEEP settlement + liability distinct

VINTAGE / LEGAL LIMIT

- UPI → instruction moving bank deposit
- DIGITAL RUPEE → direct RBI liability

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INTERFACE != CURRENCY; KEEP settlement + liability distinct.

04

STAGE 04 ONDC INTEROPERABILITY CHAIN

ONDC INTEROPERABILITY CHAIN BUYER APP OPEN PROTOCOL SELLER APP + LOGISTICS ONE GOVERNMENT MARKETPLACE

BUYER APP → OPEN PROTOCOL → SELLER APP + LOGISTICS → NETWORK != one government marketplace

| CONCEPT / ACCOUNTING BASIS | TRANSMISSION / RECONCILIATION                                   | DISTRIBUTIONAL / STABILITY LIMIT |
|----------------------------|-----------------------------------------------------------------|----------------------------------|
| BUYER APP<br>OPEN PROTOCOL | SELLER APP + LOGISTICS<br>NETWORK != one government marketplace | BUYER APP<br>OPEN PROTOCOL       |

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SELLER APP + LOGISTICS; NETWORK != one government marketplace.

05

STAGE 05 PLATFORM POWER LOOP

PLATFORM POWER LOOP MORE USERS MORE MATCHES MORE DATA BETTER RANKING SWITCHING COST ENTRY BARRIER CONTESTABILITY LEVER

MORE USERS → MORE MATCHES → MORE DATA → BETTER RANKING → SWITCHING COST → ENTRY BARRIER → INTEROPERABILITY → contestability lever

| CONCEPT / ACCOUNTING BASIS                              | TRANSMISSION / RECONCILIATION                                             | DISTRIBUTIONAL / STABILITY LIMIT                        |
|---------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------|
| MORE USERS → MORE MATCHES<br>MORE DATA → BETTER RANKING | SWITCHING COST → ENTRY BARRIER<br>INTEROPERABILITY → contestability lever | MORE USERS → MORE MATCHES<br>MORE DATA → BETTER RANKING |

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SWITCHING COST → ENTRY BARRIER; INTEROPERABILITY → contestability lever.

06

STAGE 06 DIGITAL-LENDING RESPONSIBILITY

DIGITAL-LENDING RESPONSIBILITY REGULATED ENTITY PRINCIPAL DUTY TECHNOLOGY + SERVICING ROLE DIRECT FLOW + DISCLOSURE ERASED LIABILITY

LSP / APP → technology + servicing role → DIRECT FLOW + DISCLOSURE → safeguards → OUTSOURCING != erased liability

REGULATED ENTITY → principal duty

| CONCEPT / ACCOUNTING BASIS                                                   | TRANSMISSION / RECONCILIATION                                            | DISTRIBUTIONAL / STABILITY LIMIT                                             |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------|
| REGULATED ENTITY → principal duty<br>LSP / APP → technology + servicing role | DIRECT FLOW + DISCLOSURE → safeguards<br>OUTSOURCING != erased liability | REGULATED ENTITY → principal duty<br>LSP / APP → technology + servicing role |

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LSP / APP → technology + servicing role

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SWITCHING COST → ENTRY BARRIER; INTEROPERABILITY → contestability lever.

06

## STAGE 06 DIGITAL-LENDING RESPONSIBILITY

DIGITAL-LENDING RESPONSIBILITY

REGULATED ENTITY

PRINCIPAL DUTY

TECHNOLOGY + SERVICING ROLE

DIRECT FLOW + DISCLOSURE

ERASED LIABILITY

REGULATED ENTITY → principal duty

LSP / APP → technology + servicing role

DIRECT FLOW + DISCLOSURE → safeguards

OUTSOURCING != erased liability

### CONCEPT / ACCOUNTING BASIS

- REGULATED ENTITY → principal duty
- LSP / APP → technology + servicing role

### TRANSMISSION / RECONCILIATION

- DIRECT FLOW + DISCLOSURE → safeguards
- OUTSOURCING != erased liability

### DISTRIBUTIONAL / STABILITY LIMIT

- REGULATED ENTITY → principal duty
- LSP / APP → technology + servicing role

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LSP / APP → technology + servicing role.

07

## STAGE 07 DATA-GOVERNANCE CHAIN

DATA-GOVERNANCE CHAIN

LAWFUL PURPOSE

LEGITIMATE USE

MINIMISATION + SECURITY

CORRECTION + GRIEVANCE

LAWFUL PURPOSE

CONSENT / LEGITIMATE USE

MINIMISATION + SECURITY

CORRECTION + GRIEVANCE

### CONCEPT / ACCOUNTING BASIS

- LAWFUL PURPOSE
- CONSENT / LEGITIMATE USE

### TRANSMISSION / RECONCILIATION

- MINIMISATION + SECURITY
- CORRECTION + GRIEVANCE

### DISTRIBUTIONAL / STABILITY LIMIT

- LAWFUL PURPOSE
- CONSENT / LEGITIMATE USE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONSENT / LEGITIMATE USE; MINIMISATION + SECURITY; CORRECTION + GRIEVANCE.

08

## STAGE 08 DIGITAL-ASSET TAXONOMY

DIGITAL-ASSET TAXONOMY

LEDGER ARCHITECTURE

PRIVATE TOKEN APPLICATION

SOVEREIGN RBI LIABILITY

UNIQUE TOKEN; COPYRIGHT SEPARATE

BLOCKCHAIN → ledger architecture

CRYPTO → private token application

CBDC → sovereign RBI liability

NFT → unique token; copyright separate

### MEASURE / INSTITUTION

- BLOCKCHAIN → ledger architecture
- CRYPTO → private token application

### DECISIVE DISTINCTION

- CBDC → sovereign RBI liability
- NFT → unique token; copyright separate

### VINTAGE / LEGAL LIMIT

- BLOCKCHAIN → ledger architecture
- CRYPTO → private token application

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CBDC → sovereign RBI liability; NFT → unique token; copyright separate.

09

## STAGE 09 PAYMENT-SYSTEM DISTINCTIONS

PAYMENT-SYSTEM DISTINCTIONS

MERCHANT PROCESSING FEE

INTERBANK ATM NETWORK

UPI PIN

PAYER AUTHENTICATION

NPCI SYSTEMS REMAIN FUNCTION-SPECIFIC

MDR → merchant processing fee

NFS → interbank ATM network

UPI PIN → payer authentication

NPCI systems remain function-specific

### CONCEPT / ACCOUNTING BASIS

- MDR → merchant processing fee
- NFS → interbank ATM network

### TRANSMISSION / RECONCILIATION

- UPI PIN → payer authentication
- NPCI systems remain function-specific

### DISTRIBUTIONAL / STABILITY LIMIT

- MDR → merchant processing fee
- NFS → interbank ATM network

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: UPI PIN → payer authentication; NPCI systems remain function-specific.

09

STAGE 09 PAYMENT-SYSTEM DISTINCTIONS

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INTERBANK ATM NETWORK

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CONCEPT / ACCOUNTING BASIS

- MDR → merchant processing fee
- NFS → interbank ATM network

TRANSMISSION / RECONCILIATION

- UPI PIN → payer authentication
- NPCI systems remain function-specific

DISTRIBUTIONAL / STABILITY LIMIT

- MDR → merchant processing fee
- NFS → interbank ATM network

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: UPI PIN → payer authentication; NPCI systems remain function-specific.

10

STAGE 10 PLATFORM-WORK RISK BOARD

PLATFORM-WORK RISK BOARD

MATCHING + FLEXIBILITY

ALGORITHMIC ALLOCATION

EARNINGS + DEMAND RISK

LEGAL COVERAGE

SEPARATE TOPIC 22 ROUTE

MATCHING + FLEXIBILITY

ALGORITHMIC ALLOCATION

EARNINGS + DEMAND RISK

LEGAL COVERAGE → separate Topic 22 route

CONCEPT / ACCOUNTING BASIS

- MATCHING + FLEXIBILITY
- ALGORITHMIC ALLOCATION

TRANSMISSION / RECONCILIATION

- EARNINGS + DEMAND RISK
- LEGAL COVERAGE → separate Topic 22 route

DISTRIBUTIONAL / STABILITY LIMIT

- MATCHING + FLEXIBILITY
- ALGORITHMIC ALLOCATION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EARNINGS + DEMAND RISK; LEGAL COVERAGE → separate Topic 22 route.

11

STAGE 11 DIGITAL-ECONOMY ANSWER SPINE

DIGITAL-ECONOMY ANSWER SPINE

DEFINE RAIL + ACTOR + LIABILITY

TRACE COST + SCALE + NETWORK EFFECTS

TEST COMPETITION + DATA + WORKER RISK

CONCLUDE OPEN RAILS + ACCOUNTABLE APPLICATIONS

DEFINE rail + actor + liability

TRACE cost + scale + network effects

TEST competition + data + worker risk

CONCLUDE open rails + accountable applications

CONCEPT / ACCOUNTING BASIS

- DEFINE rail + actor + liability
- TRACE cost + scale + network effects

TRANSMISSION / RECONCILIATION

- TEST competition + data + worker risk
- CONCLUDE open rails + accountable applications

DISTRIBUTIONAL / STABILITY LIMIT

- DEFINE rail + actor + liability
- TRACE cost + scale + network effects

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TEST competition + data + worker risk; CONCLUDE open rails + accountable applications.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

SERVICES PRODUCTIVITY IS MEASURED IMPERFECTLY BECAUSE QUALITY CHANGE AND

DPI SEPARATES SHARED RAILS FROM COMPETING APPLICATIONS, POTENTIALLY REDUCING

PLATFORM GOVERNANCE MUST ADDRESS SELF-PREFERENCING, TYING, DATA PORTABILITY

DIGITAL LENDING UNBUNDLES ORIGINATION, UNDERWRITING, FUNDING AND SERVICING

CBDC DESIGN CHOICES INCLUDE RETAIL VERSUS WHOLESALE, TOKEN VERSUS

DATA CAN BE NON-RIVAL YET CONCENTRATED CONTROL CREATES MARKET

RBI AND NPCI

OPTIONAL MODEL / MEASUREMENT DEPTH

- CAUTION: Services productivity is measured imperfectly because quality change and free digital
- CAUTION: DPI separates shared rails from competing applications, potentially reducing entry
- CAUTION: Platform governance must address self-preferencing, tying, data portability,
- CAUTION: Digital lending unbundles origination, underwriting, funding and servicing,

VINTAGE, BASKET OR LEGAL LIMIT

- CAUTION: CBDC design choices include retail versus wholesale, token versus account, privacy and
- CAUTION: Data can be non-rival yet concentrated control creates market power and privacy
- RBI and NPCI: RBI regulates payment systems and authorised operators; NPCI operates
- SEBI, IRDAI and PFRDA: regulate digital distribution in securities, insurance and

COMPARATIVE POLICY NUANCE

- Competition Commission of India: addresses abuse of dominance and anti-competitive
- ONDC network participants and governance bodies: implement open-protocol digital
- CAUTION: UPI sends instructions that move bank deposits; a digital-rupee transfer moves a direct
- CAUTION: ONDC allows a buyer application and seller application on different networks to

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.